Chemical heredity without genetic templates: acquisition, reproduction and retained function

Abstract

Can chemistry acquire useful functions and retain them through reproduction without genetic templates? Existing models already establish forms of template-free heredity and adaptation. We develop an event and resource account that distinguishes chemical-state copying, complementary-daughter inheritance, population renewal and cumulative functional innovation. The central example is an operated four-site stochastic population carrying three prescribed catalysts on shared local fuel. Two catalysts are prepared initially; seeding establishes a candidate before all catalyst immigration stops at a fixed time. An existing finite-stock theorem jointly guarantees both-daughter repertoire retention, continuing local conversion quotas and repeated restoration of the current population's reserves. Its three million guaranteed conversion credits comprise 1.2 million during acquisition and 1.8 million after withdrawal. We integrate the marked-process, service, recovery, conditional-composition and finite-stock arguments, while identifying the exact finite certificates they require. Comparing operating points under a common event separates recovery deadlines from the service horizon and sufficient stock ceilings from energetic efficiency. A majority-restoration example further shows why unequal hereditary events and statistical resource measures cannot be ranked by headline probabilities. The resulting framework explains what finite acquisition and retained function establish, and identifies the additional innovation, selection and compatibility obligations for a scalable theorem of successive useful adaptation. Conventional proofs, exact rational certificates and formal semigroup evidence are kept distinct. Supplied specificity, compartments and control remain explicit assumptions.

1. What must chemistry inherit?

Can chemistry accumulate adaptations without genetic templates? Specific models already give affirmative answers. The useful question is how those answers depend on the source of variation, the mechanism of reproduction, the chemical benefit and the resources that sustain them.

The compositional-genome proposal showed how mutually catalytic assemblies could transmit information through growth and splitting [1]. The subsequent GARD critique identified limitations of the dynamics it investigated, rather than a universal prohibition on chemistry before genes [2]. Later work described accumulation of adaptations through acquisition of autocatalytic cores [3], selection responses in suitable GARD ensembles [4], and compositional error-threshold behavior [5]. These results already rule out a narrative in which template-free adaptation is an untouched existence question.

Other mechanisms broaden the comparison. Inhibition-based physical models exhibit compositional heredity, spatial competition and adaptation without autocatalysis or intact compartments [6]. A recent extension studies learning of time-dependent environments [7]. Detailed comparisons with

that extension here refer to its specified preprint version, not an assumed identical final article. Experiments on vesicle populations also show a finite response to artificial selection [8]. Their measured response should not be relabelled as accumulation of new catalytic functions, just as a catalytic-acquisition theorem should not be relabelled as environmental learning.

We examine the intersection of four concrete requirements: **acquisition, reproduction, retention and useful function**. A catalyst may enter a compartment yet fail to establish. A population may retain a catalyst only because immigration never stops. Surviving descendants may contain the right species but perform little useful chemistry. Conversely, a population may keep working through repeated replacement while no particular molecular lineage persists. These distinctions determine both the biological interpretation and the mathematics needed for a credible guarantee.

Our main example is an operated stochastic population with three prescribed catalysts competing for local shared fuel [9]. Two types are initially established. A candidate is acquired during a fixed input phase; all catalyst-seed channels then stop on every realization. A complete finite-horizon result protects every required catalyst in both actual daughters at every division through the stated horizon, retains distributed service, and repeatedly restores the current population's reserves, within finite cumulative supplies. We restate that result and integrate its event composition, service, recovery and resource arguments. Its finite rational certificates are identified separately from those arguments.

The contribution is an operational synthesis, not a new claim that cumulative adaptation has been solved in this source. The example fixes a prescribed catalogue and supplied catalytic specificity. It demonstrates acquisition and hereditary maintenance of a useful repertoire, rather than generation of new catalytic designs, selection among an expanding set of alternatives, or open-ended evolution. We show exactly where these additional requirements would enter. A second quantitative example, majority restoration under continuing error [10], helps distinguish copying a label from inheriting a functional repertoire.

2. An event and resource account

2.1 Trait, preparation and descendants

A hereditary claim begins with a chemical state and a trait read from it. A trait might be a majority composition, the presence of every catalyst in a repertoire, or a function of several compartments. Let X_t denote the complete physical state, with any required observational history carried explicitly. A label w may have a set of admissible parent states $\mathcal{R}_w$. A copying statement then specifies an actual operation and an event such as

$$\{X_{\mathrm{out}}^{(1)} \in \mathcal{R}_w, X_{\mathrm{out}}^{(2)} \in \mathcal{R}_w\}.$$

The operation matters. Two complementary daughters created by partition of one parent are dependent. A retained parent and a new compartment prepared from blank molecules are a different pair of outputs. Neither can be replaced by two independent samples from a favorable marginal law.

A repertoire trait is coarser than a compositional label. For a required catalyst set J and copy floor K, define

$$\mathcal{F}_J(K) = \{x : a_{ij}(x) \geq K \text{ for every occupied site } i \text{ and every } j \in J\}. \tag{1}$$

Keeping a population in this set preserves the required types. It need not preserve their ratios, distinguish many alternative labels, or identify the molecules ancestral to a later catalyst. Those are separate observations and require separate results.

Preparation is part of the claim. A theorem from a prepared founder, a theorem from food alone, and a theorem with a supplied catalyst-seeding phase answer different questions. The finite example below uses prepared founders for two old types and supplied seeding for the candidate. Its achievement is that the expanded repertoire persists and works after seeding stops.

2.2 Success over a mission

A mission specifies the tested descendants and a horizon. For a chosen lineage of D reproduction events, a uniform ready-parent failure bound ε gives failure at most $D\varepsilon$ by conditioning successively on the exposed past. For a full binary family of depth g, the corresponding union has $2^g - 1$ reproduction events. A population-renewal mission has a different event: it tests whichever compartments occupy the population at selected stopping times. Neither event implies that one named lineage survives throughout.

Uniformity over admissible parents is essential. An estimate for an average initial parent does not automatically apply to a rare state produced after an unfavorable division. Likewise, an estimate conditional on future survival cannot simply be inserted into a proof that is supposed to establish survival. Appendix A states the conditional-composition argument used here without that circular conditioning.

2.3 What is paid?

We keep a vector of resource quantities rather than one unqualified cost. It can include initial catalysts, peak inventory, accepted feed entries, catalyst seeds, precursors, substrate conversions, division operations, time and external control. Expected consumption, a high-probability path ceiling and a physical minimum are different quantities. A sufficient feed allocation divided by a guaranteed output is a useful engineering certificate; it is not a measurement of consumption or a theorem of minimum dissipation.

External operations also belong in the account. Supplied compartments, selective catalysts, food activities, phase timing, mixing and scission can isolate a chemical-information problem without making the resulting source autonomous. Accounting for their presence is compatible with proving a substantial theorem under those assumptions.

3. A finite population that acquires and retains a catalytic repertoire

3.1 The physical count law

There are four occupied spatial sites. Each site i carries three prescribed catalyst counts $a_{ij} \in \{0, \ldots, N\}$, with $N = 2048$, and a binary shared fuel state $R_i \in \{0, 1\}$. The two old types and the candidate use the same local fuel. For a selected count a and fuel bit R, the chemical rates are:

Channel	Rate	Count and fuel update
Seed entry	$s(t)(1 - a/N)$	$a \mapsto a + 1$; fuel unchanged
Passive loss	a	$a \mapsto a - 1$; fuel unchanged
Catalytic birth	$800Ra(1 - a/N)$	$a \mapsto a + 1$; fuel becomes zero
Active loss	$400Ra$	$a \mapsto a - 1$; fuel becomes zero
Ordinary assigned conversion	$\beta Ra/N$	Count unchanged; one credit; fuel becomes zero

An empty fuel site recharges at rate β; an occupied site discharges unproductively at rate β. Each site initiates basal division at rate one, and catalyst j initiates conversion-triggered division at rate $R_i a_{ij}/N$. A trigger consumes fuel and earns its assigned conversion credit. The central coefficient is $\beta = 4{,}000{,}000$.

At division, the initiating parent and a uniformly chosen distinct victim site are replaced by two daughters. Conditional on the complete predivision state and chosen mark, independently across species,

$$K_j \sim \text{Binomial}(a_{ij}, 1/2), \qquad \text{daughter counts } K_j \text{ and } a_{ij} - K_j. \tag{2}$$

The two sisters are complementary within each species. Basal division allocates the parent's existing fuel by a complementary fair bit, independent of the complete species-split vector conditional on the predivision state and mark. Triggered division consumes it first, leaving both daughters without fuel. Victim contents are discarded and charged. Actual marked divisions whose projected state happens not to change still count as divisions.

All seeding follows the deterministic schedule

$$s(t) = 6 \quad (t < 30), \qquad s(t) = 0 \quad (t \geq 30). \tag{3}$$

No success test controls this switch. All ordinary jobs, catalytic chemistry and additional divisions continue. The physical count/fuel state is finite; cumulative observational counters are not. Total marked-event rates are uniformly bounded, so a common Poisson construction gives a nonexplosive process carrying both physical state and observations.

3.2 The preparation and operating obligations

Initially each old type has at least 512 copies at every site. Candidate counts and fuel bits may be arbitrary, including absence of the candidate everywhere. The required repertoire is the two old types before time 30 and all three types afterward. Every required type at every site must earn at least 500 credits in each applicable fixed block of length 0.1. These blocks continue during recovery and are not shifted to convenient times.

Write Ready_{512} for the event that all twelve current counts are at least 512. Let τ_0 be the first such time at or after 30. For $r = 1, \ldots, 10$, let σ_r be the first actual division strictly after τ_{r-1}, and let τ_r be the first simultaneous Ready512 time at or after the complete σ_r mark. Thus recovery concerns the current population, including any additional turnover.

3.3 The finite mission

Finite repertoire mission, restated from [9]. Under the preceding law and preparation, a single finite-stock path has all of the following properties with failure probability at most

$$\delta_* = \frac{4517975690072193}{10^{18}} < 0.004518. \tag{4}$$

1. The old types remain at least 16 at every site through time 30; the candidate is at least 16 everywhere at time 30; all three types remain at least 16 everywhere through time 60. Complete postevent states are included.

2. Every actual division in $[30, 60]$ gives both actual daughters at least 16 copies of every type. Each initiating parent therefore contains at least 32 copies of each type immediately before that division.

3. Every applicable species/site/block quota is met. Each site initiates at least five actual divisions in $[30, 60]$.

4. The eleven recoveries take at most 0.5 each, and the ten selected division waits take at most 2 each. Hence

$$\tau_{10} \le 30 + 11(1/2) + 10(2) = 55.5 < 60. \tag{5}$$

1. Through time 60 there are at most 800 million accepted feed entries,5000 accepted seed entries and 1200 actual divisions, together with the gross bounds below.

The numerical local certificates used in this restatement are specified in Appendix C. The stopped-process, service and same-path composition arguments are given in Appendices A–C. This is a consolidation of an existing conventional result, not a new cumulative-adaptation theorem.

The service total is a sum of two different obligations:

Phase	**Species/site/block quotas**	**Credits per quota**	**Guaranteed credits**
Acquisition,0–30	$2 \times 4 \times 300 = 2400$	500	1,200,000
Maintenance,30–60	$3 \times 4 \times 300 = 3600$	500	1,800,000
Full mission	6000	500	**3,000,000**

Three million is the full-mission total, not the postwithdrawal total. The renewal deadline 55.5 does not shorten the service horizon 60. These credits count modeled assigned conversion firings; the model does not calibrate a particular chemical product yield or convert a credit into a joule.

4. Why the expanded repertoire is hereditary

4.1 Both daughters, directly from one parent

On the protected path, immediately after each division the two installed counts are K_j and $a_{ij} - K_j$. Both receiving sites belong to the same population-wide floor test. Therefore both counts are at

least 16, and their sum forces the predivision parent to contain at least 32. This is a pathwise implication of the complete mark and protected event. It uses neither independent sisters nor a later recolonization of a failed daughter.

The 32-copy requirement is necessary, not sufficient, for a favorable random split. The probability argument protects all complete states; it is not an unconditional assertion that the next division succeeds from every possible 16-copy state. Nor does a Ready512 state at τ_{r-1} imply that the parent remains Ready512 until the later division.

4.2 Immigration has genuinely stopped

Fix a catalyst j and consider the face

$$Z_j = \{x : a_{ij} = 0 \text{ at every site } i\}. \tag{6}$$

After time 30, every channel that could increase a j-count either is a seed channel, whose rate is zero, or requires an existing j-catalyst. A division from a parent with zero copies has Binomial$(0, 1/2)$ allocation and gives zero to both daughters. Other species and fuel cannot create j. Thus Z_j is invariant under the postwithdrawal law, including after a finite natural-history stopping time. Appendix A gives the complete marked-path proof.

Consequently, retained candidate function depends on earlier acquired inventory and its amplification and transmission. A single empty site can still receive copies from a positive parent elsewhere; the proposition concerns absence from the whole population. It establishes a causal dependence on prior inventory, not fidelity of molecular ancestry, a many-label information capacity, or an autonomous origin of the supplied catalyst specificity.

4.3 Useful service continues through turnover

Credits are attributed to their initiating species and spatial site. Replacement does not erase previous work, and a division does not duplicate a parent's past credits into two new outputs. The fixed local quotas therefore test continuing productive service during population turnover. They are stronger than a large pooled total that could hide an inactive site, but they are different from assigning a fixed output history to a named daughter lineage.

The construction joins acquisition and retained function under one source. It does not establish that the candidate was chemically invented, that its presence was favored by a measured reproductive advantage over a competing repertoire, or that further innovations can be generated indefinitely. Those omissions identify the next interfaces rather than diminishing the stated finite theorem.

5. What the proof must control

The principal difficulty is maintaining several properties on the same history. Shared fuel couples jobs to amplification and loss. Division couples two daughters to one parent and overwrites another compartment. The law changes at time 30. A recovery estimate that suspends service, assumes a fresh supply, or restarts only from an idealized state would not prove this mission.

Three parts of the argument prevent those substitutions. First, count/fuel potentials control acquisition and all-time floors without assuming candidate readiness at time 30. Second, a credit tilt includes every other consuming channel and every division touching the tested site. Third, a recovery potential is combined with an actual resource-controlled division clock. These give uniform estimates at exposed, completed physical states. The final composition pays first failures without conditioning future dynamics on eventual global success.

Seed withdrawal needs its own generator check. Removing immigration is not automatically favorable for a nonmonotone recovery potential, and a certificate proved with seeds cannot simply be reused. The source checks the actual seed-free rows while retaining the same physical division law and its positive operator. Likewise, resource tails are proved for the switched law itself; they are not inferred by a pathwise monotonicity claim between seeded and seed-free populations.

The resulting failure allocation is

$$\frac{9545923}{9062500000} + \frac{11}{100000} + 10e^{-8} + 3e^{-20} \leq \delta_*. \tag{7}$$

The first term protects acquisition, every required floor, all fixed-grid service and the per-site division count. Eleven local recovery allowances and ten waiting allowances follow. Three global resource tails are charged once. No independence between these events is asserted.

6. What the resource comparison actually says

6.1 A same-path finite supply certificate

The effective source has intrinsic capacity $4 \times 3 \times 2048 = 24576$ catalyst slots. The prepared old inventory is separate from later inputs. It is at most 16384 copies when the candidate starts absent, or 24576 when its initial counts are arbitrary. At most four initially occupied fuel tokens are supplied.

Cumulative quantity through 60	**Sufficient ceiling**
Accepted feed/recharge entries	800,000,000
Consumed fuel events	800,000,004
Accepted catalyst seeds	5000
Catalytic-birth plus seed precursor entries	800,005,004
Actual divisions/compartment handling operations	1200
Discarded/restored catalyst slots	7,372,800

Handling-slot restoration supplies capacity, not catalyst molecules. All new catalysts still enter by seeds before withdrawal or paid catalytic birth. Declared job substrates can be debited once per conversion; their sufficient allocation is bounded by the consuming-fuel ceiling under that convention. This is effective-model accounting, not an elemental mass balance for unspecified chemical structures or a price for scission work.

The feed-cap/output-floor ratio is

$$\frac{800{,}000{,}000}{3{,}000{,}000} = \frac{800}{3} \simeq 266.7. \tag{8}$$

It compares a sufficient ceiling with a guaranteed floor. Using only the 1.8 million postwithdrawal credits in the denominator would answer a different question. Neither ratio is measured efficiency, minimum fuel demand or dissipated energy.

6.2 Comparing operating points under one declared event

The continuously seeded comparison family in [9] has complete parameter points at beta = 8,000,000, beta = 4,500,000 and beta = 4,000,000. The stronger beta = 8,000,000 recovery event with allowance 0.2 is contained in the event with allowance 0.5. Thus all three points can be compared against a common specification: the original fixed-grid service through 60, eleven recoveries of at most 0.5, ten waits of at most 2, and completion of the selected renewal sequence by 55.5. This event containment does not change any physical rate or replace the original stronger beta = 8,000,000 statement.

Common continuously seeded specification	**beta = 8,000,000**	**beta = 4,500,000**	**beta = 4,000,000**
Source recovery allowance	0.2	0.5	0.5
Source renewal deadline	52.2	55.5	55.5
Credits by 60	3,000,000	3,000,000	3,000,000
Accepted-feed ceiling	1,558,500,000	900,000,000	800,000,000
Feed ceiling/guaranteed credits	519.5	300	266.7
Source failure ceiling	<0.006	$\leq 0.005464632462485986$	$\leq 0.004517975690072193$

The beta = 4,000,000 seed-withdrawal theorem additionally removes all catalyst immigration after 30 at its stated guarantee. That switch is an added result, not an unmentioned feature of every row in this comparison table. The table shows certified operating tradeoffs under an explicit common event; it does not prove optimality or stochastic ordering in beta. Different certificates can produce a smaller failure bound at a lower rate without proving that lowering the rate always improves the physical process.

6.3 Restoration is a different resource claim

In the finite-affinity restoration source [10], each compartment has $n = 100m$ molecules in two conformations. A parent with at most 15% wrong molecules contacts a racemic blank compartment for time 3, then both outputs restore separately for time 20. Spontaneous errors continue. The quantitative bounds below assume integer $m \geq 10$, intensive rates $a = \alpha = 1$, fixed-in-time $9/200 \leq r \leq 11/200$ and $9/1000 \leq \kappa \leq 11/1000$, and a binomial racemic blank independent of the fixed ready parent. Appendix D prints the exact channel law and derives the bound uniformly on this box. The outputs are a retained parent and a new copy, not two daughters from partition of a growing vesicle. The source explicitly uses conformational templating during contact; the absence of genetic sequence copying does not mean absence of every templating interaction.

Its complete one-cycle bound is

$$\varepsilon_m = e^{-3m/10} + e^{-8m} + 2\rho_m, \qquad \rho_m \le 40(5/8)^m. \tag{9}$$

At $m = 40$, a sharper finite calculation gives $\varepsilon_{40} \le 10^{-5}$, so 100 selected-lineage cycles succeed with probability at least 0.999. The simpler estimate $2e^{-0.003n}$ alone would not justify that example. Appendix D gives the channel-specific restoration and noisy-parent copying proofs, the full parameter-box inequalities, the finite comparison and the conditioning argument.

Expected gross fuelled firings are at most $15.4n$ per cycle, including both directions; expected spontaneous flips are $46\kappa n$. At the nominal $\kappa = 0.01$ and $n = 4000$, the latter is 1840. Two output compartments contain 8000 molecules in total. These quantities cannot be ranked directly against the repertoire mission's high-probability feed ceiling: the traits, outputs, operation and statistical resource quantities differ. Keeping their units and events visible resolves the apparent comparison without discarding either result.

7. From retained function to successive useful innovation

The finite mission makes one important bridge explicit: an acquired catalyst need not remain dependent on continued catalyst immigration. It also makes the unproved bridges visible. To claim successive useful adaptation under fixed rules, one must additionally specify how variants arise, why their chemical effects are useful, how selection establishes them, and how previous functions survive while the next useful change occurs.

The comparison with earlier work is therefore mechanistic. Autocatalytic-core acquisition [3] already offers a route to accumulated adaptations. Inhibition-based state changes and temporal learning [6, 7] already demonstrate other adaptive capabilities. A new finite-budget theorem must identify a specific additional event and prove it. A larger repertoire of possible states, an improved productivity score, and accumulation of independently specified catalytic jobs are not interchangeable endpoints.

A scalable target can be stated without asking a fixed closed finite system for unlimited novelty. Fix generative chemical rules and a declared compartment/reproduction mechanism. For acquisition count r and failure tolerance η, specify what can scale: molecular inventory, population, duration, fuel/material throughput, chemical diversity and rule-description length. Starting from a declared unadapted preparation, require useful variants to arise through the allowed chemistry, establish under a specified selection mechanism, retain earlier functions, and permit further beneficial descendants. Charge supplied catalyst specificity, designed substrate catalogues and environmental schedules rather than hiding complexity in the initial conditions.

One can then ask a precise question: which resources must grow to make the joint event reliable? A family redesigned for every r remains a meaningful intermediate construction if identified honestly. Continuing inputs can support a long-running open system. A finite-stock mission has a finite horizon. Neither finite success nor a large state repertoire alone proves open-ended evolution.

7.1. Measurements that distinguish the claims

The event definitions suggest a small set of measurements and controls. These are proposed consequences of the accounting, not reports of new experiments.

Observation or control	What it tests	What still needs a separate demonstration
Stop catalyst entry while maintaining declared food/fuel inputs	Whether productive continuity requires ongoing catalyst immigration	Whether unmeasured background chemistry or contamination can regenerate the catalyst
Measure both daughters immediately after one complete division	Transmission from one parent before later recovery or recolonization	Long-term retention, useful activity and subsequent descendant production
Measure later current-population copy reserves separately	Recovery after turnover under the actual operating law	Survival of a particular daughter or molecular lineage
Count new substrate conversions per site and time block	Continuing local function, rather than persistent accumulated product	A selective advantage over an alternative repertoire
Compare genuinely new variants with appropriate competitors under the same conditions	Chemical benefit and selection acting on variation	Retention of earlier functions while further innovations arise

The invariant-face argument makes one conditional inference exact. If all production of an absent catalyst is excluded by the declared generator after withdrawal, later catalyst-dependent service implies that the entire population has not previously lost that catalyst. This is a causal inventory criterion, not a universal experimental inference from a positive readout. Background synthesis, unobserved immigration or a different carrier invalidates its premise. The criteria therefore identify both what to measure and which mechanism assumptions need independent support.

8. Evidence, applicability and the scope of this synthesis

The repertoire result is conventional mathematics with exact rational certificate support. Its marked-process construction, conditional restart and finite-stock coupling are analytical arguments. A historical Lean acquisition/grid theorem belongs to its original source; it is not full formal verification of the switched finite-stock mission. The restoration source has source-matched formal results for its finite-state semigroups and discrete mission construction, with conventional identification of sample paths and reaction-event compensators. These are different, useful forms of evidence.

The effective repertoire law assumes prescribed catalysts and jobs, supplied compartments, controlled immigration withdrawal and paid but externally implemented division/replacement. The large rate and supply scales are sufficient conditions, not demonstrated physical necessities. No laboratory calibration, endogenous membrane-work budget or commercial efficiency claim follows. A complete elementary chemical realization would need to justify those interfaces rather than rename them as autonomous chemistry.

The present synthesis concentrates on a tractable core rather than treating every historical manuscript as a necessary publication. The exact proof arguments that can be integrated are given below. The central reset proof now uses a fixed rational table and an explicit finite row certificate, without an exact low-band construction premise. Its required comparisons are printed and supported by source-matched historical checks; no newly packaged computation has been run here. The separate operating-comparison dependencies, including the legacy beta8m branching source, remain preserved with the original companion. Supplement S1 supplies the exact data, checking-source editions, saved historical outputs and retained comparison proof sources, with attribution, licence notices and derivative mappings. Keeping a search or optimization engine

private is compatible with releasing a complete proof; keeping an indispensable witness private is not.

The central conclusion is affirmative and limited: within a declared operated chemical model, acquisition can be followed by reproduction, retained catalyst function and repeated population renewal after catalyst input stops, under finite resources. The stronger question of successive chemically generated useful innovations under a fixed scalable mechanism remains a target to formulate and test against existing answers.

Reproducibility, source attribution and evidence

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These computations were not rerun during preparation. Historical source/output agreement supports the specified conventional finite certificate; it is not a fresh execution receipt. The whole archival program needs parts of its former provenance environment and is not advertised as a turnkey portable verifier. No new Lean compilation or full switched-process formalization is claimed. The beta8m comparison retains its original Cert/Defs proof source and pinned external dependencies; optional fresh formal checking is distinct from disclosing those proof sources.

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References

[1] Segré, D., Ben-Eli, D. and Lancet, D. (2000). Compositional genomes: Prebiotic information transfer in mutually catalytic noncovalent assemblies. PNAS 97,4112–4117. DOI.

[2] Vasas, V., Szathmáry, E. and Santos, M. (2010). Lack of evolvability in self-sustaining autocatalytic networks constraints metabolism-first scenarios for the origin of life. PNAS 107,1470–1475. DOI.

[3] Vasas, V., Fernando, C., Santos, M., Kauffman, S. and Szathmáry, E. (2012). Evolution before genes. Biology Direct 7,1. Full article.

[4] Markovitch, O. and Lancet, D. (2012). Excess mutual catalysis is required for effective evolvability. Artificial Life 18,243–266. DOI.

[5] Gross, R., Fouxon, I., Lancet, D. and Markovitch, O. (2014). Quasispecies in population of compositional assemblies. BMC Evolutionary Biology 14,265. DOI.

[6] Bunin, G. and Rivoire, O. (2025). Evolutionary features in a minimal physical system: Diversity, selection, growth, inheritance, and adaptation. PNAS 122, e2425753122. DOI. Detailed author-version comparison: arXiv 2411.03843 v 2,10 July 2025; published and preprint versions are not presumed identical.

[7] Rivoire, O. and Bunin, G. (2026). Learning to outgrow the competition: Reaction–diffusion systems that adapt to time-dependent environments. PNAS 123, e2605428123. DOI. Detailed claims here refer to arXiv 2512.01007 v 2,14 August 2026; final-publication metadata and abstract are distinct evidence from the inspected preprint text.

[8] Sokolskyi, T. and Baum, D. (2026). Prebiotically Plausible Vesicle Populations Can Respond to Selection for Greater Turbidity via Emergent Cooperative Dynamics. Langmuir 42,17787–17795. DOI.

[9] Template-free heredity of an acquired catalytic repertoire. P161, research source collection (author field unspecified in the source),2026. Source manuscript, version4, deterministic seed withdrawal,39-page PDF. Exact source SHA 256 `731042cdc927f7af7d6c11d9d6f559f8be66944d5e9d61bcc5e511e346b8624f`; PDF SHA 256 `66a723c07e6dbbe6ff9aa7427b19604ea0d3c2db0103de4fb2c06d55b8044ff3`. Source Theorem 1 and Appendices A–H; the source identity, preserved bibliography and text edition are recorded in Supplement S1. Complete source edition in Supplement S1: P161 source (Supplement S1), including G.1-G.4 and its supporting appendices. These are retained dependencies of the operating comparison; the source and its retained comparison support are supplied together in Supplement S1.

[10] Reliable chemical inheritance with continual errors at finite affinity. P118, research source collection (author field unspecified in the source),2026. Source manuscript P118, dated 5 October 2026. Main source SHA 256 `e2532ee3239290bffb9a4d09af06cde5b65fd74f347ff2cf68bcce7d31eec796`; PDF SHA 256 `bd9bc6035f6043575c686f6aeeafe4d7f966f8b977bec6d95245959e4bd72992`. Theorems 3.1-3.2, Corollary 3.4, proof and sample-path appendices. The conventional quantitative argument is integrated in Appendix D; exact source attribution is retained. Source edition: P118 source (Supplement S1).

Appendix A. Physical inheritance, restart and whole-mission composition

A.1. Complementary daughters and the population floor

Proposition A1. In the source of Section 3, fix an interval I and required set J. Suppose every complete state in I belongs to $\mathcal{F}_J(16)$. At every actual division in I, both daughters contain at least 16 copies of every $j \in J$, directly from the initiating parent. The parent's predivision count is at least 32 for each such type.

Proof. The complete division mark installs K_j at the parent site and $a_{ij}(t-) - K_j$ at the distinct victim site. Both are counts tested by the same postmark population floor. Their sum is the parent's premark inventory. No intervening seed or catalytic-birth mark occurs within this complete update.

This proves both statements. The argument is pathwise and makes no independence claim. It also shows why a division from a parent with 16–31 copies cannot lie on the protected path. □

Proposition A2. Under the seed-free post 30 law, the face Z_j in (6) is invariant, including after any almost-surely finite stopping time of the natural marked history at which the state belongs to Z_j.

Proof. At a state in Z_j, seed birth has zero rate. Every other chemical channel that changes a j-count or earns its conversion credit has a factor a_{ij} and thus zero rate. Other-species reactions and fuel recharge/discharge cannot create j. At division, Binomial$(0, 1/2)$ is zero almost surely, so both daughters and all untouched sites still have zero copies. Every positive-rate complete mark therefore maps Z_j into itself. Bounded rates imply finitely many marks on each bounded interval. Induction over marks proves invariance there; a countable union extends the conclusion to all later times. Restart at a finite natural-history stopping time gives the conditional version by A.2 below. The face need not be a single absorbing physical state, because other types and fuel can still evolve. □

A.2. The marked law at exposed stopping times

Choose a uniform rate exceeding the total marked-event rate at every physical state in either phase. In the present source,93515864 suffices. Independent exponential waits at this rate and independent marks implement each physical channel, its victim and full complementary allocation, with a residual dummy proposal. Carry cumulative observations with the state. A dummy proposal is not material input or division; a state-null actual division remains a counted physical mark.

At deterministic times, Poisson independent increments and the exponential lack of memory give the same future construction from the completed present state. This extends to grid-valued stopping times by splitting over their possible values. For a finite stopping time σ, approximate from above by $\sigma_m = 2^{-m}\lceil 2^m\sigma\rceil$. On bounded intervals only finitely many proposals occur almost surely, so the right-continuous state at σ_m eventually equals the completed state at σ. Future cylinder laws converge; bounded convergence and a monotone-class argument extend the identity to bounded future-path functionals. Localization removes a bound on σ.

Consequently a pointwise local estimate valid from every admissible completed state applies conditionally on the complete past at such a start. The filtration reveals no future wait or mark. For post 30 starts, the restarted physical law is homogeneous and seed-free. Service-grid position, cumulative stock counters and previous success flags remain carried history; they are not replenished.

For a finite physical projection, put $P = I + L/\Lambda$. Its Poisson-mixture law is $p_t = e^{-\Lambda t}\sum_{n\geq 0}(\Lambda t)^n p_0 P^n/n!$. The series converges absolutely and solves $p_t' = p_t L$ with initial value p_0; uniqueness of the finite linear system identifies it with $p_0 e^{tL}$. This supplies the finite-semigroup reading used by the restoration example. Observation counters remain marked-path quantities rather than being falsely declared finite-state. For a bounded-rate set of channels, the firing count minus the integral of its instantaneous rate is an integrable compensated-count martingale on finite horizons. Taking expectations identifies the expected count with that rate integral, including actual marked firings whose physical projection is unchanged.

This is a conventional marked-process construction and proof. It does not claim a newly compiled Lean path-space theorem.

A.3. First failures rather than future-success conditioning

Let G protect acquisition, all required floors, fixed-grid service and initiating-division counts through 60. Suppose $\mathbb{P}(G^c) \le a$. At every admissible post 30 recovery start, suppose

$$\mathbb{P}(H > h, D > h \mid \mathcal{F}_\sigma) \le b, \tag{A1}$$

where H is first simultaneous Ready512 and D first floor exit in the freshly stopped local run. At every finite exposed start suppose the next actual division wait exceeds w with conditional probability at most c. Define the eleven recoveries and ten waits as in Section 3.

Charge each first failed episode only on histories where earlier deadlines succeeded and no earlier floor failure occurred. That prefix indicator is measurable at the exposed start. On G, a late recovery implies the local event $\{H > h, D > h\}$. Conditional expectation of its prefix-weighted indicator is therefore at most b times the probability of that prefix. Sum the eleven terms. Apply the same argument to the ten waits. Adding G^c gives

$$\mathbb{P}(G^c \text{ or a renewal deadline fails}) \le a + 11b + 10c. \tag{A2}$$

This does not condition the future law on G. Successful preceding deadlines keep all relevant starts inside the protected horizon when $30+11h+10w \le 60$. Four rate-one basal clocks ensure $c \le e^{-4w}$, since the first actual division is no later than their next firing, even while triggers and chemistry continue. At $h = 1/2, w = 2$, the deadline is 55.5 and the wait contribution is $10e^{-8}$.

A.4. Transfer to finite supplies

Construct the unlimited and finite-stock versions of the **same switched law** with identical proposal waits, channel decisions, victims and complete split marks. In the finite version stop at the first proposal that would exceed a declared cumulative allocation. Before that proposal the states, rates and ledgers agree by induction over proposals. On an event bounding every cumulative use through 60 by the allocation, no such excess occurs, so the entire histories agree through 60.

If the three resource-cap failures have total probability at most $3e^{-20}$ on the unlimited switched law, combining this coupling with(A2) costs that probability once. No local recovery receives a new budget. This proves the event-level finite-stock transfer rather than merely appending stock numbers to an unlimited-source theorem.

Appendix B. Acquisition, all-time floors and continuing service

B.1. The precise certificate needed for establishment

For $N = 2048$ and $\beta = 4{,}000{,}000$, define

$$h_5(a) = \frac{1}{(a+1)(a+2)(a+3)(a+4)(a+5)}, \quad C_0 = 1860480 h_5(12)(4/3)^{12},$$

$$A(a)=\begin{cases}C_0(3/4)^a, & a\le 12,\\ 1860480h_5(a), & a\ge 13,\end{cases}\qquad g(0)=-A(0)/2,\quad g(a)=A(a)\min(6a,70)\ (a\ge 1),$$

$$W(a,1)=A(a),\qquad W(a,0)=A(a)+g(a)/\beta. \tag{B1}$$

The finite rational branching certificate proves $0\le W\le 80$, $W\ge 1$ below 16, and $W\le 6/10^8$ above 512. For $\Phi_j=\sum_i W(a_{ij},R_i)$, its required full-generator bounds are

$$L\Phi_j\le -\tfrac{1}{2}\Phi_j+4b_0\quad (t<30),\qquad L\Phi_j\le 4b_0\quad (t\ge 30,\ \text{all } j\text{-counts}\ge 16),\quad b_0=13/10^7. \tag{B2}$$

The second assertion is only a live-region retention bound. It is not an all-state seed-free acquisition theorem. Both include the complete physical division and victim law. The concrete finite row/moment obligations are identified in Appendix C.4; neither acquisition probability nor eventual mission success is assumed as an input.

The relation of the finite row check to the physical law is as follows. Write $u=a/N$, let $U_f\in[0,2]$ be the other two species' total normalized count, and let $v\in[0,1843200]$ be their combined active birth/death intensity. Their ordinary/active fuel loss is $H_f=\beta U_f+v$. Put $E_r(a)=\mathbb{E}[W(K,r)]$, $K\sim\text{Binomial}(a,1/2)$. With seed coefficient $s\in\{0,6\}$, the literal branching upper operator tested by the certificate is

$$\begin{aligned}\mathcal{S}_sW(a,0)&=s(1-u)[W(a+1,0)-W(a,0)]+a[W(a-1,0)-W(a,0)]\\&\quad+\beta[W(a,1)-W(a,0)]+2E_0(a)-2W(a,0),\\ \mathcal{S}_sW(a,1)&=s(1-u)[W(a+1,1)-W(a,1)]+a[W(a-1,1)-W(a,1)]\\&\quad+800a(1-u)[W(a+1,0)-W(a,1)]+400a[W(a-1,0)-W(a,1)]\\&\quad+[\beta(1+u)+H_f][W(a,0)-W(a,1)]\\&\quad+E_1(a)+E_0(a)-2W(a,1)+(u+U_f)[2E_0(a)-W(a,1)].\end{aligned} \tag{B9}$$

At $a=0$ the predecessor can be defined as zero because its coefficient vanishes; at $a=N$ the upward coefficients vanish. All binomial outcomes, including below-floor daughters, enter E_r. A basal division gives complementary sisters with the same marginal count law and one inherited fuel token when the parent is fueled; a triggered division consumes that token and has two fuel-zero outputs. Summing basal marks over the four possible parents gives two daughter gains, total parent loss $-\Phi_j$, and uniform-victim loss $-\Phi_j$: the latter follows because each site is a victim for three other parents with probability 1/3. This is the basal $-2W$ in (B9). For triggers, dropping the nonpositive victim-weight loss gives precisely its last upper-bound term. The derivation uses complementary marginals, never sister independence.

For clarity the full daughter expectations in (B9) have short exact formulas. The identity

$$h_5(k)=\frac{1}{24}\int_0^1 t^k(1-t)^4dt$$

follows by integrating the degree-four polynomial, or the beta integral. Averaging t^K gives $((1+t)/2)^a$; substitution $x=(1+t)/2$ and subtraction of the integral over $[0,1/2]$ yield

$$J(a) := \mathbb{E}h_5(K) = 32h_5(a) - \frac{4}{3}\sum_{j=0}^{4} \frac{(-1)^j \binom{4}{j}}{(a+j+1)2^{a+j+1}}.$$

Hence, writing $D = 1860480$,

$$E_A = DJ(a) + 2^{-a} \sum_{k=0}^{\min(a,12)} \binom{a}{k} [A(k) - Dh_5(k)],$$

$$E_g = 70E_A + 2^{-a} \sum_{k=0}^{\min(a,11)} \binom{a}{k} [g(k) - 70A(k)], \qquad E_1 = E_A, \quad E_0 = E_A + E_g/\beta. \tag{B10}$$

The finite corrections are exact because the geometric prefix ends at 12 and $g(k) = 70A(k)$ from 12 onward. Thus a few corroborating direct sums are not being substituted for a full-moment identity. Each residual in (B9) is affine in U_f, v; verifying its four rectangle corners covers every possible foreign load, including correlated loads along the actual path.

For the fixed $\beta = 4000000$ core, the independent literal source enumerates all $2049 \cdot 2 \cdot 4 = 16392$ seed-on rows of $\mathcal{S}_6 W + W/2$, and all $2033 \cdot 2 \cdot 4 = 16264$ live seed-free rows. The latter actually checks the stronger $\mathcal{S}_0 W + W/2 \le b_0$; dropping its negative term proves the weaker postwithdrawal inequality in (B2). Removing seeds increases this decreasing potential's drift, so a seed-on pass alone cannot establish the live seed-free result. The source's finite inequalities and their literal row lists remain indispensable certificate content; no new execution of them is asserted here.

The shape properties of (B1) also have an elementary all-count proof. The ratio $A(a+1)/A(a)$ is $3/4$ through the prefix and $(a+1)/(a+6)$ from 12 onward. The unfueled 0-to-1 ratio at beta one million is $6000036/7999996 < 1$; the varying prefix obeys $3(M+6)/(4M) < 1$ for $M \ge 1000000$, after which its multiplier is constant. Both branches are positive and decreasing, $A(15) = 1$, and $C_0 < 80$. Direct rational substitution at 512 gives the prepared bound. For larger beta the unfueled branch is a convex combination of its beta-one-million value and A, so these bounds persist. These shape arguments do not assume survival or acquisition.

Stop the two-old-species potential at its first complete floor failure. Its initial value is at most $8(6/10^8)$, its drift before exit is at most $8b_0$ across both phases, and its stopped value is at least one on a failure. Dynkin's formula gives

$$\mathbb{P}(\text{old floor failure by60}) \le 48/10^8 + 480b_0. \tag{B3}$$

No post-exit drift premise is used. For the candidate, $\Phi_j(0) \le 320$, so the seed-on inequality and an integrating factor yield

$$\mathbb{E}\Phi_j(30) \le 320e^{-15} + 8b_0(1 - e^{-15}).$$

From the actual time 30 distribution, stop at candidate floor failure. If a count is already below 16 at 30, the stopping time is 30 itself and the barrier already applies. Another 30 time units cost at most $120b_0$. Thus acquisition plus subsequent candidate-floor failure has probability at most $320e^{-15} + 128b_0$. Adding(B3) gives

$$48/10^8 + 320e^{-15} + 608b_0. \tag{B4}$$

This is why no candidate Ready512 assumption is imposed at 30.

B.2. A service inequality on the full shared-fuel law

Fix a site/species pair. Let Q count its ordinary conversions since an exposed block start, without being overwritten at division. Let D be the first complete mark violating the current required count floors. For $0 < z \le \rho \le 1$, use the killed exponential weight

$$Y = z^Q \rho^R.$$

Set $H_s = 800 \cdot 3N \cdot 9/16 + 3\beta + 8$. When $R = 0$, recharge contributes $-\beta(1-\rho)z^Q$. Incoming division either keeps fuel empty or supplies fuel, preserving or decreasing this weight. Seeds and passive deaths leave it unchanged.

When $R = 1$, the selected ordinary job contributes $\beta(a/N)(z-\rho)z^Q$. Discharge and the two other ordinary jobs cost at most $3\beta(1-\rho)z^Q$. Each active birth/loss pair has rate $800Nu(3/2-u) \le 800N(9/16)$; three such pairs give the first part of H_s. Own division intensity is at most 4. Incoming victim replacement has intensity at most 4, so all touching divisions cost at most $8(1-\rho)z^Q$. This bound includes the full donor/sister fuel mark. Non-touching events preserve the function.

If the required floor is K, then $a \ge K$ before D, and $z - \rho \le 0$. Therefore the two scalar conditions

$$\theta \le \beta(1-\rho), \qquad \beta(K/N)(z-\rho) + H_s(1-\rho) + \theta\rho \le 0 \tag{B5}$$

imply $L^D Y \le -\theta Y$. Killing replaces a nonnegative postmark weight by zero and cannot increase this generator. The resulting supermartingale estimate is

$$\mathbb{E}[1_{\{D>w\}} Y_w] \le e^{-\theta w} Y_0 \le e^{-\theta w}.$$

On $Q_w < m$, the conservative lower bound $Y_w \ge \rho z^m$ holds. Hence

$$\mathbb{P}(D > w, Q_w < m \mid \text{exposed past}) \le \rho^{-1} z^{-m} e^{-\theta w}. \tag{B6}$$

This is an intersection bound, not a probability conditional on future floor survival. Division-coupled conversion credits can only help the actual quota. Seed withdrawal contributes zero to the generator of this count-independent weight, so the argument is unchanged in the two phases.

At $z = 3/4, \rho = 79967/80000, \theta = 1650, K = 16$, recharge gives equality in the first condition, and the second residual is $-598067/10000$. A rational lower Taylor polynomial for e^{165} gives

$$\frac{12}{\rho}(4/3)^{500} e^{-165} < 10^{-8}. \tag{B7}$$

This charges at most twelve coordinate quotas per block. Applying(B6) with past-measurable live-prefix indicators and then event inclusion charges all 600 original blocks by 600×10^{-8}, without changing any block during recovery.

For an entirely rational version of(B7), put $P_{200}(x) = \sum_{j=0}^{200} x^j/j!$. The finite comparison is

$$\frac{12}{\rho}\frac{(4/3)^{500}}{P_{200}(165)} < \frac{1948303823}{250000000000000000} < 10^{-8}.$$

Since $P_{200}(165) \le e^{165}$, this implies the stated probability bound.

B.3. Closing the base operating event

Each site's actual initiation count dominates its rate-one basal count on $[30, 60]$. For a Poisson mean 30 count, exponential Markov with factor 6 gives at most $6^4 e^{-25}$ for fewer than five firings. Four sites cost at most $5184e^{-25}$, without independence from floors or service.

Adding this to(B4) and(B7), then using $e^{15} > 2900000$ and $e^{25} > 10^{10}$, proves the direct base-event bound. Conservatively replace $608b_0$ by $720b_0$ to retain the accepted allocation

$$a_* = 48/10^8 + 320/2900000 + 600\{(6/5)b_0 + 10^{-8}\} + 5184/10^{10} = 9545923/9062500000. \quad \text{(B8)}$$

This event includes candidate establishment, every required floor and grid quota, and each site's five initiating divisions. The spare $112b_0$ is a conservative allocation, not a separate unproved hazard.

Appendix C. Recovery, resource clocks and finite certificates

C.1. A general potential-and-clock bound

Consider a finite physical marked process stopped on leaving a live set. Reset marks may have null physical updates and are still counted. Write Q_0 for the live generator with reset gains removed and resets treated as killing, and B for the positive live-poststate reset gain. Suppose

$$U \ge 1, \quad U(x) \le M \text{ at admissible starts}, \quad Q_0 U \le -\gamma U, \quad BU \le CU. \quad \text{(C1)}$$

Let ν be the reset intensity. A bounded physical statistic F, retained at the terminal stopped mark, is assumed to satisfy

$$0 \le F \le F_{\max}, \quad \nu - a \le -\kappa L F, \quad |\Delta F| \le b, \quad \sum_e r_e(x)(\Delta_e F)^2 \le V. \quad \text{(C2)}$$

For $T > 0, K \ge 1, \delta > \kappa F_{\max}$ and $0 < \theta < 3/b$, the surviving probability is at most

$$Me^{-\gamma T}\sum_{j=0}^{K-1}\frac{(CT)^j}{j!} + \mathbb{P}\{\text{Poisson}(aT+\delta) \ge K\} + \exp\left[-\theta(\delta/\kappa - F_{\max}) + \frac{VT\theta^2}{2(1-b\theta/3)}\right]. \quad \text{(C3)}$$

Proof. Let $P_j(t)$ be the unnormalized live law with exactly j resets. Its equations are $P_0' = P_0 Q_0$ and $P_j' = P_j Q_0 + P_{j-1} B$. Apply(C1), an integrating factor and induction to obtain $P_j(t)1 \le P_j(t)U \le Me^{-\gamma t}(Ct)^j/j!$. This bounds histories with fewer than K resets.

The compensated resource $M_t^F = F_t - F_0 - \int_0^t LF_s\,ds$ gives

$$\Lambda_T := \int_0^T \nu_s\,ds \le aT + \kappa(F_{\max} + M_T^F).$$

For bounded jumps, the exponential integral remainder and positive Taylor series give $e^{\theta x} - 1 - \theta x \le x^2\theta^2/[2(1-b\theta/3)]$ for $|x| \le b$. The compensated-jump exponential supermartingale therefore bounds the probability $\Lambda_T > aT + \delta$ by the third term of(C3).

Finally realize the reset count as $N_T = Y(\Lambda_T)$ for a unit-rate Poisson primitive Y. Although Y and its clock may be dependent,

$$\{N_T \ge K, \Lambda_T \le aT + \delta\} \subseteq \{Y(aT + \delta) \ge K\}$$

holds pathwise because every Poisson path is nondecreasing. This proves the middle term without assuming clock/count independence. Sum the three events. The restart argument of A.2 gives the same bound conditionally at every admissible completed stopping-time start. □

C.2. The actual fuel clock

For this source take $F = \sum_i R_i$ and $Z = \sum_i R_i \sum_j a_{ij}/N$. The actual total division intensity is $\nu = 4 + Z$, with $Z \le 3F$. Counting the complete marks gives

$$LF = \beta(4 - 2F - Z) - A_{\text{active}} - D_{\text{fuel}}, \qquad A_{\text{active}} \ge 400NZ.$$

Consequently, for $\kappa = 3/(5\beta + 1200N)$ and $a = 4 + 4\beta\kappa$,

$$-\kappa LF \ge \kappa\{(\beta + 400N)Z + 2\beta F - 4\beta\} \ge Z - 4\beta\kappa = \nu - a.$$

Fuel lies between zero and four. A complete division can remove at most two fuel tokens, so the jump bound is 2. Recharge, discharge and ordinary jobs contribute at most 20β to the squared-jump rate; active birth/loss contributes at most $5400N$; divisions contribute at most $16 \times 4 = 64$. Thus $V = 20\beta + 5400N + 64$. These are statewise bounds, valid with arbitrary nuisance histories. Seeds change no fuel and do not alter them.

C.3. The recovery potential and its numerical use

Let $H^+[a][r]$ be the companion's literal two-column rational profile export, with column 0 unfueled and column 1 fueled. For $16 \le a < 512$, define explicitly

$$A_{\text{rec}}(a) = H^+[a][0], \qquad \Delta(a) = H^+[a][0] - H^+[a][1].$$

Set $A_{\text{rec}}(a) = \Delta(a) = 0$ below 16. At and above 512 use the exact formulas

$$f(a) = \frac{(980 - a)(981 - a)(982 - a)}{(a - 10)(a - 9)(a - 8)} \quad (a \le 768), \qquad f(a) = f(768)(10/11)^{a-768} \quad (a > 768),$$

$$A_{\mathrm{rec}}(a) = f(a)/f(512), \qquad \Delta(a) = A_{\mathrm{rec}}(a)\{258 + k(a)\}/8000000.$$

For $512 \le a < 768$, $k(a) = 6(1 - a/N)[f(a+1)/f(a) - 1] + a[f(a-1)/f(a) - 1]$; for $a \ge 768$, $k(a) = 5647a/56320 - 6/11$. Set both entries zero below 16 and define

$$h(a,0) = A_{\mathrm{rec}}(a), \qquad h(a,1) = A_{\mathrm{rec}}(a) - (11/8)\Delta(a), \qquad U = \sum_{i,j} h(a_{ij}, R_i). \tag{C4}$$

Finite rational checks establish the barrier, positivity and fuel order; the seed-free chemical inequalities are $(G_0 + 150)h \le 0$ below 512 and $(G_0 + 150)h \le 1/4700$ above it. They include counts 16,511,512,768 and 2048. The positive division operator is unchanged by withdrawing chemical seeds. The direct fixed-data certificate below gives $BU \le 478U$, with its finite comparisons stated as verification obligations and supported by source-matched historical checks. Since $U \ge 1$ on a live nonReady state and removal of resets contributes at least 4 killing,

$$Q_0 U \le -(154 - 12/4700)U, \qquad U \le M = 338971453059/125000 \quad \text{at all admissible starts.} \tag{C5}$$

The recovery live domain is the set where all 12 current counts are at least 16 and at least one is at most 511. Stop on the first floor failure or simultaneous Ready512; the terminal complete mark and its actual fuel bits define that stopping event. The inequality $U \ge 1$ is needed only on this live, nonReady domain, not on all count states.

Here is the link from the checked chemical rows to the full population generator. Let $s(a) = 6(1 - a/N)$, $m(a) = a$, $b(a) = 800a(1 - a/N)$, $d(a) = 400a$. With $\sigma = 0$ after withdrawal, the reference two-fuel chemical operator is

$$\begin{aligned}
G_\sigma h(a,0) &= \sigma(1 - a/N)[h(a+1,0) - h(a,0)] + m(a)[h(a-1,0) - h(a,0)] + \beta[h(a,1) - h(a,0)],\\
G_\sigma h(a,1) &= \sigma(1 - a/N)[h(a+1,1) - h(a,1)] + m(a)[h(a-1,1) - h(a,1)]\\
&\quad + b(a)[h(a+1,0) - h(a,1)] + d(a)[h(a-1,0) - h(a,1)]\\
&\quad + [\beta(3 + a/N) + 800N][h(a,0) - h(a,1)].
\end{aligned} \tag{C9}$$

The selected coordinate's own ordinary conversion and discharge contribute $\beta(1 + a/N)$. Each foreign coordinate's ordinary/active fuel consumption is bounded by $T(z) = 800z(1 - z/N) + 400z + \beta z/N \le T(N) = 400N + \beta$, since $\beta \ge 400N$ makes T increasing on $[0, N]$. This proves the final reference coefficient in (C9). The actual fuel-off rate is no larger; multiplying their difference by $h(a,0) - h(a,1) \ge 0$ shows that the actual chemical generator is bounded above by the reference generator on each coordinate. Its sum is valid with shared fuel and correlated foreign histories. No independent-coordinate process replaces the population.

C.3.1. A direct certificate for the literal recovery profile

The central reset bound can be proved without constructing an unknown exact low-count profile. Write $A = A_{\mathrm{rec}}$, $D = \Delta$, $t_0 = A$, $t_1 = A - D$, and set $\delta = 10^{-18}$, $\ell = 9997/10000$, $c = 11/8$. Use the literal two-column rational table $P(a,r) = H^+[a][r]$ as a proof envelope and put

$$X(a,r)=\max\{0,P(a,r)-\delta\},\qquad P(a,r)-\delta\le t_r(a)\le P(a,r)\quad(512\le a\le 2048,\ r\in\{0,1\}). \tag{C10a}$$

Only the fixed low entries define the new witness there; its high values remain the exact formula(C4). The high table is used only after the explicit finite enclosure comparisons in(C10a), not as an unexplained predecessor witness. These comparisons, and integrality of each $10^{18}P(a,r)$, are checked in the retained historical independent full-mission source. No band solution is needed to state or verify them. The data file SHA256 is `1cbf7592037531b0c61ca9c5c01f40e418bf5932b291c25273490e535fb12542`.

The first finite obligation is

$$A(a)>0,\qquad 0\le D(a)\le A(a)/1251\quad(16\le a<512). \tag{C10b}$$

It consists of explicit rational comparisons on 496 fixed pairs. Its nontrivial upper comparison is exactly $\ell t_1\le h(\cdot,1)$, because $h(a,1)-\ell t_1(a)=[3A(a)-3753D(a)]/10000$. The source's later seed-free comparison checks use the literal low entries here; they do not require the old exact-profile enclosure to establish this particular inequality.

The corresponding tail comparison has an elementary proof. For $512\le a<768$, the positive rational f decreases and

$$\frac{f(a-1)}{f(a)}=\frac{983-a}{980-a}\,\frac{a-8}{a-11}.$$

The seed term in $k(a)$ belongs to $[-6,0]$. The passive term is nonnegative and is less than $768[(203/200)(503/500)-1]=1619712/100000<17$, using $980-a>200$ and $a-11>500$. Thus $-6\le k<17$. For $768\le a\le 2048$, the printed linear formula and $1/10<5647/56320<1/9$ give $0<k<228$. Consequently

$$0<D(a)/A(a)<1/10000\quad(512\le a\le 2048).$$

Together with(C10b), this proves positivity, fuel-zero domination and

$$A(a)\le P(a,0),\qquad \ell X(a,r)\le h(a,r),\qquad 0<h(a,1)\le h(a,0) \tag{C10c}$$

on every live coordinate row. In the tail, use(C10a) and $X\le t\le P$; below 512, use $P=t$, $X\le t$ and(C10b). The positive comparison is a property of the literal witness and its tail, not of an unobserved predecessor profile.

For $a<32$ put $M^+(a)=0$. Otherwise define the complete paired-floor moment and daughter residual by

$$\begin{aligned} M^+(a)&=2^{-a}\sum_{k=16}^{a-16}\binom{a}{k}P(k,0),\\ p(a,r)&=1+r(2+a/N),\\ Y(a,r)&=2p(a,r)M^+(a),\\ d_*&=93867565097/200000000. \end{aligned} \tag{C10}$$

The remaining reset certificate is the following two finite all-row comparisons:

$$\begin{aligned}\max_{16\le a\le 511,\ r\in\{0,1\}}[Y(a,r)-d_*X(a,r)] &\le b_L=-147104211175367700201/500000000000000000,\\ \max_{16\le a\le 2048,\ r\in\{0,1\}}[Y(a,r)-d_*X(a,r)] &\le b_A=3343277526704540851/1250000000000000000.\end{aligned}\tag{C11}$$

These are outward upper bounds, not equalities identifying the true maxima. Their signed combination is exactly

$$b_L+11b_A=-367902757/500000000000000000<0.\tag{C11a}$$

For independent verification, read the fixed rational P, check its grid integrality and(C10a)-(C10b), and evaluate(C10) for every $a=32,\dots,2048$. Start a summation at $\binom{a}{16}$, and update consecutive coefficients by $\binom{a}{k+1}=\binom{a}{k}(a-k)/(k+1)$; include every k through $a-16$. Compare every one of the 4066 count/fuel residuals with b_A, and the 992 deficient rows with b_L. There are 2,035,153 summand iterations. No profile search, linear-system solve, later cap stage or population-state enumeration occurs in this finite specification. A checksum, two-number table or unexecuted program label alone is not a verification of these comparisons.

Here is the conventional soundness proof. A fixed retained site has three other possible parents. Each has rate at most four and avoids that site as victim with probability 2/3, so the total retained contribution is at most $8U$. For a chosen parental coordinate (a,r), the actual parent's rate is at most $p(a,r)$. On a surviving split, the sisters have counts $K,a-K$, with $16\le K\le a-16$. Fuel-zero domination and(C10c), followed by complementary-binomial symmetry, bound their combined expected potential by $2M^+(a)$. This uses the two marginals of one complementary pair, without assuming independent sisters or independent coordinate dynamics. Dropping other coordinates' floor tests and simultaneous-Ready killing only increases the nonnegative gain. Therefore

$$BU\le 8U+\sum_{i,j}Y(a_{ij},R_i).$$

Every live nonReady prestate has a deficient coordinate. Apply the first bound in(C11) there and the second at the other eleven coordinates. By(C11a) and(C10c), $\sum_{i,j}Y(a_{ij},R_i)\le d_*\sum_{i,j}X(a_{ij},R_i)\le(d_*/\ell)U$. It follows that

$$\begin{aligned}BU&\le C_{\mathrm{direct}}U<478U,\\ C_{\mathrm{direct}}&=8+d_*/\ell=95467085097/199940000,\\ 478-C_{\mathrm{direct}}&=104234903/199940000>0.\end{aligned}\tag{C12}$$

The retained term remains $8U$; it is not inflated by an old/new-profile transfer. The complete physical marks, shared fuel and state-null divisions are retained. Withdrawing chemical seeds changes neither this reset law nor this proof. We continue to price recovery with the conservative constant 478, leaving all advertised mission bounds unchanged.

C.3.2. Finite-proof evidence and its boundary

Equations(C10a)-(C10b),(C11), including grid integrality, remain explicit finite verification obligations. The source-matched historical q1 check reports exactly the first-stage coefficient and outward

bounds used here. Its rounded output is identified by the source's export-identity record as the same literal low table. The later independent full-mission checker directly reads the complete literal table, checks its high-tail enclosure and grid integrality, and recomputes the same clipped-weight, full paired-floor residuals without a band solve. Its saved result and stdout report the same two upper endpoints in(C11). This provides additional historical finite-computation evidence for precisely the literal-envelope version printed here. The cap2048 rows are a time-independent finite lemma; their use here does not extend the historical q1 probability bound from one duration to another.

The q1 source has SHA256 `73acc0c43cee75b6c3b5ef495fc5139e029023e6d97500a56855164d149b537b`, and its constants have SHA256 `bd7a5b38df771cce413bec381176f0c068a2946b44b3176a16f76aa4f003b829`. The later independent literal-row checker has SHA256 `47159a820c0b653fb90e8f0b05b6941f86266cfdc9ef67568f6b3b3faef807af`. Its report, source, result and stdout are supplied as identified archival editions in Supplement S1; the provenance index distinguishes original hashes from edition hashes. Their historical recomputation is distinct from a new execution result. The original q1 raw frozen directory was not found in the source-linked locations checked; that archive limitation is not evidence that its mathematical lemma has no proof. The certificate's exact comparisons are not erased or replaced by an unspecified reset assumption.

The old exact low-count band construction and its enclosure are no longer logical premises of this central beta4m reset step. The finite comparisons have not been re-evaluated in preparing this successor, and no fresh portable-checker acceptance is claimed. Public use must retain the complete independently inspectable data/formulas/finite verification route or exact sufficient accessible proof support, with appropriately scoped validation claims.

C.3.3. Separate comparison dependencies and recovery pricing

The beta8m/0.2 comparison retains its own old-profile chemical, enclosure and mission dependencies. Its legacy branching support `cert0` and `Defs.lean` is supplied in Supplement S1 under `comparison_support`, with the exact source/dependency pins and package-specific rights index. The original constructors and identified archival checker editions are supplied; incidental path redactions are mapped and do not alter mathematical expressions. This central reset simplification does not newly validate that comparison, the beta4.5m row or the remaining chemical, acquisition, service and finite-stock obligations. No citation is removed while its proof is still used.

Use(C3) with $T = 1/2, K = 14, \delta = 1/100, \theta = 1/640, C = 478$. At beta = 4,000,000, the actual clock exponent is 28228801/460320 and its comparison mean is 1080309/350900. The conservative exponent 9141301/460320 and mean 321/100 suffice. Positive lower Taylor polynomials and upper geometric remainders yield

$$q_{\mathrm{off}}(1/2) \le 8622092430140/10^{18} < 1/100000. \tag{C6}$$

This covers words with 0–13 resets and the event of at least 14 resets. It bounds $\mathbb{P}(H > 1/2, D > 1/2)$, not recovery failure without paying floor exits. The source's beta = 8,000,000 calculation instead uses $T = 1/5, K = 9$ and obtains a bound below 1/50000. Neither parameter statement is substituted for the other.

The scalar numerical claim in(C6) can be checked using rational expressions alone. Put $P_{-}(x) = \sum_{j=0}^{200} x^j/j!$, and, for $0 \le x < 202$, put

$$P_+(x) = P_-(x) + \frac{x^{201}}{201!}\frac{1}{1 - x/202}.$$

The remaining positive exponential tail has successive-term ratio at most $x/202$, so $P_-(x) \le e^x \le P_+(x)$. With $\mu = 321/100$ and $E = 9141301/460320$, an explicit rational upper bound in(C6) is

$$\frac{M \sum_{j=0}^{13} (478/2)^j/j!}{P_-(\gamma/2)} + 1 - \frac{\sum_{j=0}^{13} \mu^j/j!}{P_+(\mu)} + \frac{1}{P_-(E)}.$$

The source compares this sum with $8622092430140/10^{18}$ by exact arithmetic before a final outward ceiling. No separately rounded component sum is used to assert that smaller ceiling.

C.4. The essential finite-certificate obligations

The following are source-specific finite proofs, rather than assumptions of the desired mission:

Obligation	**Concrete objects and finite checks**
Branching and retention	The explicit function(B1); full binomial daughter moments; all 2049 counts, two fuels and four load corners for the seeded bound; live counts 16–2048 for the seed-free bound
Literal recovery input	Exact rational profile values below 512; positive tail(C4); barriers, maximum and fuel order;4066 physical count/fuel rows for seed-free chemistry
Paired reset	For $K \sim \text{Binomial}(a, 1/2)$, exact truncated moment of the profile over $16 \le K \le a - 16$; retained-victim coefficient 8; deficient-coordinate max inequality over all count/fuel rows
Scalar bounds	Taylor-polynomial/rational inequalities for service, recovery, waits and stock caps; outward rounding only after summing where specified

The paired-reset support cannot be replaced by two independent daughter probabilities. The direct proof in C.3 uses the literal-table envelope and the fully specified finite certificate; no exact low-band construction is a premise of this central reset step. Its grid, tail-enclosure, low-gap and complete residual comparisons remain verification obligations, supported by the retained historical independent record. An unexecuted newly packaged checker is not thereby a fresh pass. The data and sufficient finite-proof support must be available with any public argument. Other operating comparisons retain their separate old-profile dependencies.

C.5. Feed, seeds and divisions

At an occupied fuel site, the combined consuming rate is at most

$$C_* = 4\beta + 1200N + 3 = 18457603.$$

Indeed each of three species contributes $800Nu(3/2 - u) + (\beta + 1)u$, increasing through $u = 1$ at this beta; add discharge β. Basal victim-fuel loss has intensity F; triggered victim loss is at most $3F$.

Let A_T count accepted recharges. For $t \geq s \geq 0$, tilting only these marks by e^t and using $v = e^{-sF}$ gives

$$L_t v/v \leq \beta(4-F)(e^{t-s}-1) + F(C_* + 1 + 3e^s)(e^s - 1) \leq 4k(t,s),$$

where k is the maximum of the two coefficients. In a triggered division with occupied victim fuel, split $e^{2s} - 1 = (e^s - 1) + e^s(e^s - 1)$: the parent token is charged in C_*, the extra victim token in the last term. Thus

$$\mathbb{P}(A_{60} > B) \leq \exp\{s(4 - F_0) + 240k(t,s) - tB\}. \tag{C7}$$

Choose $t = 1/10000, s = 400/22457607, B = 800000000$. The elementary $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$ bounds the two coefficients by explicit rationals. Cross multiplication shows that the first dominates:

$$k(t,s) \leq \max\left\{\frac{\beta(t-s)}{1-t+s}, \left(C_* + 1 + \frac{3}{1-s}\right)\frac{s}{1-s}\right\} = \frac{\beta(t-s)}{1-t+s}.$$

Substitution in(C7) therefore bounds the exponent by

$$-550899147934266425 1200/5043026607980323551 < -1092 < -20.$$

Seeds preserve A and F, so their withdrawal does not alter this proof. Seed intensity is at most 72 before 30 and zero afterward. With exponential parameter $1/10$ and $e^{1/10} < 10/9$, the 5000-seed tail is below e^{-260}. Actual division intensity is at most 16 throughout; parameter $1/5$ and $e^{1/5} - 1 < 31/140$ give the 1200-division tail at most $e^{-192/7} < e^{-20}$. State-dependent accepted counts need not be independent Poisson variables for these exponential moment bounds.

Finally complete-mark conservation gives

$$F_0 + A = F + B_{\mathrm{cat}} + H + Q + U_{\mathrm{waste}} + W_R, \quad B_{\mathrm{cat}} + H + Q + U_{\mathrm{waste}} \leq A + 4,$$

$$B_{\mathrm{cat}} + E \leq A + 4 + E, \qquad W_R \leq D, \quad W_S \leq 3ND. \tag{C8}$$

These identities and inequalities hold at every prefix, including terminal jumps. Use A.4 to transfer the complete unlimited-source mission on the simultaneous cap event. Substituting(B8) and(C6) into(A2) and charging three global tails proves (7), with a rational outward bound(4).

Appendix D. Majority restoration with continual errors

This appendix integrates the conventional argument of [10], including its physical count law, uniform parameter box, return potentials, noisy-parent copying estimate and finite witness. These are established results of that source, not new mathematics claimed for this synthesis. The retained parent and new copy are the two outputs; this protocol does not describe vesicle division. It uses conformational templating during contact and therefore addresses heredity without genetic sequences, rather than absence of all templating interactions.

D.1. The exact law and operated cycle

A compartment contains n molecules in two conformations. Fix either label, let y be the number wrong relative to it and put $k = n - y$. Fuel and waste are chemostatted, with reverse factor $r = e^{-A}$; A is the affinity in units of $k_B T$. The concentration is fixed as n varies. With intensive restoration rate $a = 1$, the wrong-count birth and death rates are

$$u_n(y) = \frac{ky(y-1)}{n^2} + r\frac{k(k-1)(k-2)}{n^2} + \kappa k, \qquad d_n(y) = \frac{yk(k-1)}{n^2} + r\frac{y(y-1)(y-2)}{n^2} + \kappa y. \quad \text{(D1)}$$

The three terms in each direction are respectively forward cooperative conversion consuming fuel, reverse conversion releasing fuel, and spontaneous error. Falling factorials are exact, including at the boundary; a zero factor makes its channel zero. In particular $u_n(0) > 0$. Purity is not absorbing, and the argument never conditions on absence of errors. Exchanging the labels gives $u_n(n-y) = d_n(y)$.

For copying, write the wrong counts as (p, z) for parent and new compartment. Parent jumps have rates $\kappa(n-p), \kappa p$. New-compartment jumps have spontaneous rates $\kappa(n-z), \kappa z$, together with copying rates

$$\begin{array}{c|cc} & \text{forward} & \text{reverse} \\ z \longmapsto z+1 & \alpha(n-z)p/n & r\alpha(n-z)(n-p)/n \\ z \longmapsto z-1 & \alpha z(n-p)/n & r\alpha z p/n. \end{array} \quad \text{(D2)}$$

Only parent molecules catalyze the copying reactions. Cooperative restoration is off during contact; spontaneous errors continue in both compartments. No operation reads the label or freezes the parent.

Take $n = 100m$, integer $m \geq 10$, and fixed parameters

$$\alpha = a = 1, \qquad \frac{9}{200} \leq r \leq \frac{11}{200}, \qquad \frac{9}{1000} \leq \kappa \leq \frac{11}{1000}. \quad \text{(D3)}$$

Start from any parent with $y_0 \leq 15m$ and a new compartment whose wrong count is Binomial$(n, 1/2)$. Run (D2) for $T_c = 3$, then separate the actual pair and restore each under (D1) for $T_r = 20$. Each output is ready when its wrong count is at most $15m$. Subsequent restoration laws are independent conditional on the pair at separation; the copying counts themselves need not be independent. The proof below uses only their marginal capture bounds and a union bound.

For either finite count chain use $Lf(x) = \sum_c w_c(x)[f(\tau_c x) - f(x)]$. The finite law and time-dependent expectation identity follow by A.2, or directly by differentiating the finite matrix exponential. All forthcoming probability estimates are at specified fixed times.

D.2. An unstopped two-potential return bound

Let $V, W \geq 0$, $\beta_1, \beta, \delta \geq 0$, and $\mu, v_S, v_H, w_R > 0$. Suppose

$$LV \leq \beta_1, \quad LW \leq -\mu W + \beta + \delta \mathbf{1}_S, \quad S \subseteq \{V \geq v_S\}, \quad \text{Bad} \subseteq \{V \geq v_H\} \cup \{W \geq w_R\}.$$

For every initial state x_0 and $T \geq 0$,

$$\mathbb{P}_{x_0}(X_T \in \mathrm{Bad}) \le \frac{V(x_0)+\beta_1 T}{v_H} + \frac{e^{-\mu T}W(x_0) + [\beta + \delta(V(x_0)+\beta_1 T)/v_S]/\mu}{w_R}. \tag{D4}$$

Indeed $\mathbb{E}V(X_t) \le V(x_0) + \beta_1 t \le V(x_0) + \beta_1 T$. Markov's inequality therefore bounds $\mathbb{P}(X_t \in S)$ by this last expression divided by v_S. Insert that bound into the expectation inequality for W, multiply by $e^{\mu t}$, and integrate. Dropping the factor $1 - e^{-\mu T} \le 1$ gives the numerator in (D4). Apply Markov's inequality to V, W on the displayed covering of Bad. The chain has not been stopped at an excursion above a barrier.

For integers $0 \le L < R < I < H \le n$, $L+1 < H$, and slopes $q, q_1 > 1$, set

$$V(y) = q_1^{(\min(y,H)-L)^+}, \qquad W(y) = \begin{cases} q^{(y-L)^+}, & y < H, \\ 0, & y \ge H. \end{cases} \tag{D5}$$

Suppose on $L < y < H$ that $u_n \le d_n/q_1$ and $(q-1)(u_n - d_n/q) \le -\lambda$, and suppose $u_n(y) \le U$ for $y \le L$, $d_n(H) \le U$. Then

$$LV \le (q_1 - 1)U, \qquad LW \le -\lambda W + (q-1)U + \lambda + Uq^{H-1-L}\mathbf{1}_{\{y=H\}}. \tag{D6}$$

For V, the generator is zero below L and above H. At L it is $u_n(L)(q_1 - 1)$; between the endpoints it is $q_1^{y-1-L}(q_1-1)(q_1 u_n - d_n) \le 0$; at H its downward contribution is nonpositive. For W, the generator is zero below L and above H, is $u_n(L)(q-1)$ at L, and equals $W(q-1)(u_n - d_n/q)$ on the interior up to $H-2$. At $H-1$, replacing the upward target value by zero only decreases this generator. At H, the sole contribution is $d_n(H)q^{H-1-L}$. The added constant $(q-1)U + \lambda$ covers the plateau where $W = 1$. This proves every boundary case in (D6).

The exceptional re-entry set $\{H\}$ is contained in $\{V \ge q_1^{H-L}\}$, and $\{y > R\} \subseteq \{V \ge q_1^{H-L}\} \cup \{W \ge q^{R+1-L}\}$. Hence from every $y_0 \le I$, (D4) gives

$$\mathbb{P}_{y_0}(Y_T > R) \le \pi + \frac{e^{-\lambda T}q^{I-L} + [(q-1)U + \lambda + Uq^{H-1-L}\pi]/\lambda}{q^{R+1-L}}, \quad \pi = \frac{q_1^{I-L} + (q_1-1)UT}{q_1^{H-L}}. \tag{D7}$$

The different slopes pay for returns from above the barrier. With equal slopes the corresponding factor would grow as q^{I-R}; the source's choice gives $q^{23}q_1^{-10} < 1$ per unit m. This is a feature of this proof, not a universal obstruction to other return arguments.

D.3. Exact inequalities on the entire parameter box

Choose $(L, R, I, H) = (10m, 15m, 28m, 38m)$, $q = 11/10$, $q_1 = 27/20$, $\lambda = n/1000$, $U = 33n/100$. We give the algebraic certificate for (D6), rather than invoking sampled drift values.

Write $N = n, Y = y, K = N - Y$. For either slope $q_* \in \{11/10, 27/20\}$, the quantity $N^2(d_n/q_* - u_n)$ is affine in r, κ. Its coefficients are $Y(Y-1)(Y-2)/q_* - K(K-1)(K-2)$ and $(Y/q_* - K)N^2$. Both are negative on $N/10 \le Y \le 19N/50$, $N \ge 1000$: here $100 \le Y < K$, so the nonnegative falling factorial is strictly smaller for Y than for K, and $q_* > 1$. The minimum is therefore at $r_1 = 11/200, \kappa_1 = 11/1000$. At that vertex, expand

$$N^2(d_n/q_* - u_n) = H_{q_*} + A_{q_*} + B_{q_*},$$

$$H_{q_*} = \frac{YK^2 + r_1Y^3 + \kappa_1 N^2 Y}{q_*} - (Y^2K + r_1K^3 + \kappa_1 N^2 K),$$

$$A_{q_*} = YK(1 - 1/q_*) + 3r_1(K^2 - Y^2/q_*) \geq 0, \qquad B_{q_*} = 2r_1(Y/q_* - K) \geq -2r_1N. \tag{D8}$$

Put $s = Y - N/10$, $t = 19N/50 - Y$. Then $s, t \geq 0$ and $N = 25(s+t)/7$. The following table gives the exact coefficients of $H_{q_*} - (c + 10^{-4})N^3$; the identity is checked by substituting for Y, N and collecting the four monomials.

(q_*, c)	s^3	s^2t	st^2	t^3
$(11/10, 1/100)$	5439579/6036800	7877943/1207360	1905857/241472	61505/241472
$(27/20, 0)$	2684653/14817600	12605801/2963520	3549799/592704	15845/197568

Every entry is positive. Since $10^{-4}N^3 \geq 2r_1N$ for $N \geq 1000$, (D8) proves $d_n/q_* - u_n \geq cN$. For $q_* = 11/10$, multiplication by $q_* - 1 = 1/10$ gives the required contraction $N/1000$; for $q_* = 27/20$ it proves the escape inequality. Finally each direction's forward rate is at most $n/4$, its reverse rate at most rn, and its spontaneous rate at most κn. Thus both rates are at most $(1/4 + r + \kappa)n \leq 0.316n < U$, proving the caps. These are exact polynomial inequalities over the whole box.

With $T = 20$, (D7) consequently becomes

$$\pi_m = \frac{q_1^{18m} + 231m}{q_1^{28m}}, \qquad \rho_m = \pi_m + \frac{e^{-2m}q^{18m} + 34 + 330q^{28m-1}\pi_m}{q^{5m+1}}. \tag{D9}$$

Every captured state $y_0 \leq 28m$ has restoration failure at most ρ_m, uniformly in (D3).

D.4. Copying while the parent fluctuates

For the copying pair let $v = z - n/2$, $w = p - n/2$, $\Lambda = \alpha(1 + r) + 2\kappa$, $\varrho = \alpha(1 - r)$. The daughter up/down rates from (D2), including spontaneous errors, are $(n/2 - v)(\Lambda/2 + \varrho w/n)$ and $(n/2 + v)(\Lambda/2 - \varrho w/n)$. Thus the exact mean drifts are $Lw = -2\kappa w$, $Lv = -\Lambda v + \varrho w$.

For $0 < \theta \leq 1$, set $G_s = \exp\{\theta[A(s)v + B(s)w] + \varphi(s)\}$, with $A' = \Lambda A$, $B' = 2\kappa B - \varrho A$. Put $P(x) = e^x - 1 - x$, $P_-(x) = e^{-x} - 1 + x$. For nonnegative x, $0 \leq P_-(x) \leq P(x)$. Expanding the jump exponentials, their linear terms cancel the time derivative exactly. The total daughter rate is at most $n(\alpha + \kappa)$, because the copying sum is at most αn for $0 < r < 1$; the parent rate is κn. Consequently

$$\frac{(\partial_s + L)G_s}{G_s} \leq \varphi'(s) + n[(\alpha + \kappa)P(\theta A) + \kappa P(\theta B)]. \tag{D10}$$

For $0 \leq x \leq \theta \leq 1$, the power series gives $P(x) \leq x^2[1/2 + \theta \sum_{j \geq 3} 1/j!] \leq c_\theta x^2$, where $c_\theta = 1/2 + 2\theta/9$. To justify the last rational bound, $j! \geq 6 \cdot 4^{j-3}$ for $j \geq 3$, so the sum is at most

$(1/6)/(1-1/4) = 2/9$. If $0 \le A \le 1$, $0 \le B \le B_{\max} \le 1$, choose $\varphi' = -n\theta^2 c_\theta[(\alpha+\kappa)A^2 + \kappa B_{\max}^2]$. Equation (D10) and the finite expectation identity give $\mathbb{E}G_T \le \mathbb{E}G_0$.

For a daughter tail at time T, choose

$$A(s) = e^{-\Lambda(T-s)}, \quad B(s) = B_0[e^{-2\kappa(T-s)} - e^{-\Lambda(T-s)}], \quad B_0 = \frac{1-r}{1+r}.$$

These solve the backward equations, obey $A(T) = 1, B(T) = 0$, $0 \le B \le B_0 < 1$, and $B(0) = h(T) = B_0(e^{-2\kappa T} - e^{-\Lambda T})$. Initially $p = y_0$ is deterministic and z is binomial. Its exponential moment is $\cosh(\theta A(0)/2)^n \le \exp(n\theta^2 A(0)^2/8)$; the inequality follows by integrating $\tanh x \le x$. Since $\int_0^T A(s)^2 ds \le 1/(2\Lambda)$, exponential Markov's inequality proves

$$\mathbb{P}(z_T > I) \le \exp\left\{-\theta(I+1-n/2) + \theta h(T)(y_0 - n/2) + n\theta^2\left[c_\theta\left(\frac{\alpha+\kappa}{2\Lambda} + \kappa B_0^2 T\right) + \frac{e^{-2\Lambda T}}{8}\right]\right\}. \quad \text{(D11)}$$

The integer threshold is $I+1$. For the parent choose $A = 0$, $B(s) = e^{-2\kappa(T-s)}$, $B_{\max} = 1$ instead. The same argument yields

$$\mathbb{P}(p_T > I) \le \exp\{-\theta(I+1-n/2) + \theta e^{-2\kappa T}(y_0 - n/2) + n\theta^2 c_\theta \kappa T\}. \quad \text{(D12)}$$

The parent remains a random part of the joint chain throughout this calculation. In particular the κB^2 charge is not removed by treating it as a deterministic source; no independent-daughter-molecule law conditioned on a fluctuating parent is assumed.

Here are sufficient rational specializations of these bounds at $T = 3$. Over (D3), $e^{-6\kappa} \ge 117/125$, $e^{-3\Lambda} \le 413/10000 < 1/4$, $B_0 \ge 189/211$. Hence $h(3) \ge (189/211)(117/125 - 413/10000) \ge 4/5$. For completeness the first exponential bound follows from $(1 - 33/32000)^{64} \ge 117/125$ and $e^{-x} \ge 1 - x$ at $x = 33/32000$. For the second, $3\Lambda \ge 3189/1000$ and $e^{3189/1000} \ge (27182818283/10^{10})^3[1 + 189/2000 + (189/2000)^2/2]^2 \ge 2425/100$. The lower bound on e follows from $\sum_{j=0}^{13} 1/j! > 27182818283/10^{10}$; all remaining comparisons here are rational cross multiplications.

For the daughter put $\theta = 1/8$, so $c_\theta = 19/36$. The bounds $(1+\kappa)/(2\Lambda) \le 1011/2126 < 12/25$, $\kappa B_0^2 T \le 33/1000$, and $e^{-2\Lambda T}/8 \le 1/128$ show that the exponent in (D11), for $y_0 \le .15n, I = .28n$, is at most

$$-\frac{3n}{400} - \frac{1}{8} + \frac{n}{64}\left[\frac{19}{36}\left(\frac{12}{25} + \frac{33}{1000}\right) + \frac{1}{128}\right] = -\frac{3n}{400} - \frac{1}{8} + \frac{4457n}{1024000} \le -\frac{3n}{1000}.$$

For the parent put $\theta = 1, c_\theta = 13/18$. Its exponent in (D12) is at most

$$\left[\frac{11}{50} - \frac{7}{20}\frac{117}{125} + \frac{13}{18}\frac{33}{1000}\right] n - 1 = -\frac{2513n}{30000} - 1 \le -\frac{2n}{25}.$$

Thus, for all parameters and all ready parents under consideration,

$$\mathbb{P}(z_3 > 28m) \le e^{-3m/10}, \qquad \mathbb{P}(p_3 > 28m) \le e^{-8m}. \quad \text{(D13)}$$

These deliberately sufficient algebraic bounds recover the source's capture rates without needing its sharper intermediate variance estimates.

D.5. Exponential scale, the finite witness and repeated cycles

Apply (D9) after separation conditional on the actual pair. If a count is captured its return failure is at most ρ_m; otherwise charge its failure to (D13). A union bound proves the one-cycle, two-output estimate

$$\varepsilon_m = e^{-3m/10} + e^{-8m} + 2\rho_m. \tag{D14}$$

No independence between the two copying errors is used. Reflection of (D1), the paired copying channels and the racemic initial law proves the same statement for either inherited label.

To give the decay and the finite witness explicitly, let $u = (20/27)^m$, $v = (10/11)^m$, $y = (11/10)^m$, $E = e^{-2m}$. Expanding (D9) gives

$$\rho_m = u^{10} + 231mu^{28} + \frac{10}{11}Ey^{13} + \frac{340}{11}v^5 + \frac{33000}{121}(y^{23}u^{10} + 231my^{23}u^{28}). \tag{D15}$$

Here the superscripts on u, v, y are ordinary powers. The following exact rational comparisons prove $\rho_m \le 40(5/8)^m$ for every integer $m \ge 10$. Set $a_0 = 5/8$. Bound $m \le 2^m$. The six bases in (D15), after absorbing that factor where needed and dividing by a_0, are bounded as follows:

Term without coefficient	**Upper bound on normalized base**	**Upper bound on its m-th power for $m \ge 10$**
u^{10}	$1/10$	$1/10^8$
mu^{28}	$1/10$	$1/10^8$
Ey^{13}	$3/4$	$3/50$
v^5	$1987/2000$	$937/1000$
$y^{23}u^{10}$	$357/500$	$69/2000$
$my^{23}u^{28}$	$1/10$	$1/10^8$

For example the first normalized base is $(20/27)^{10}/a_0$, the second is $2(20/27)^{28}/a_0$, and the last is $2(11/10)^{23}(20/27)^{28}/a_0$. For the third, use $e^{-2} \le 13534/100000$, following from the preceding lower bound on e. Each base is at most one, so its m-th power is at most its tenth power; the right column is an outward rational bound for that tenth power. Weighting the column by the coefficients in (D15) gives

$$\frac{1}{10^8} + \frac{231}{10^8} + \frac{10}{11}\frac{3}{50} + \frac{340}{11}\frac{937}{1000} + \frac{33000}{121}\left(\frac{69}{2000} + \frac{231}{10^8}\right) < 40. \tag{D16}$$

This supplies the claimed decay using a short list of exact power comparisons, without a fitted asymptotic estimate. For $m \ge 27$, the simpler source bound also follows. Indeed $e^{-3/10} \ge (1 - 3/640)^{64} \ge 74/100$, $e^{-8} \le 1/15$, so $e^{-8m} \le (1/1000)e^{-3m/10}$. Also $(5/8)/(74/100) \le 845/1000$ and $(845/1000)^{27} \le 107/10000$. Therefore

$$\varepsilon_m \le \left(1 + \frac{1}{1000} + 80\frac{107}{10000}\right)e^{-3m/10} < 2e^{-3m/10} = 2e^{-0.003n}. \tag{D17}$$

At $m = 40$ retain (D14) rather than weakening it to (D17):

$$\varepsilon_{40} \le e^{-12} + e^{-320} + 80(5/8)^{40} \le \frac{1}{162754} + \frac{1}{10^{10}} + \frac{80}{10^8} < 10^{-5}. \tag{D18}$$

The exponential comparisons can be reduced to finite rational statements: $\sum_{j=0}^{64} 12^j/j! \ge 162754$, $320^5/5! > 10^{10}$, and $10^8 5^{40} \le 8^{40}$. Thus the quoted hundred-cycle witness has a complete conventional finite proof; the displayed $2e^{-12}$ bound alone would yield only $1 - 200e^{-12}$, about 0.998771.

The bound is uniform over every ready parent. Conditional on each successful cycle and the selected ready output, the next failure probability is at most ε_m. Iterating conditional expectation, or summing first failures, gives selected-lineage success at least $1 - g\varepsilon_m$ over g cycles. The formal interface uses a state-based ready-preserving function; actual daughter selectors satisfy that hypothesis. This manuscript makes no enlarged formal claim about arbitrary history-dependent controllers. A full binary family charges $2^g - 1$ cycles instead. In particular (D18) gives at least 0.999 for 100 selected-lineage cycles at $n = 4000$. This is a finite mission, not open-ended evolution.

D.6. What its resource estimate measures

The counting convention is expected gross fuelled reaction events, not a finite-stock high-probability ceiling. During copying the combined forward-and-reverse fuelled rate is $\alpha[(n-z)(p+r(n-p)) + z((n-p)+rp)]/n \le \alpha n = n$, since each parent-dependent bracket is at most n. This gives at most $3n$ expected gross events. In a restoring compartment the total forward rate is $ky(n-2)/n^2 \le n/4$; the sum of reverse rates is at most rn. Charging both forward and reverse events therefore costs at most $(1/4 + 11/200)n < .31n$ per unit time. Two compartments restored for 20 time units contribute at most $12.4n$. The whole cycle is bounded by $15.4n$ expected gross events, including both directions in both phases. This is not a pathwise fuel inventory. Net fuel consumption is bounded by the gross count, so the expected chemical work supplied by these chemostats is at most $A\,15.4n$. The apparatus, supplied blank compartment, switching and chemostats remain physical assumptions; this is not their total energetic price.

At all times there are two compartments with total spontaneous-flip intensity $2\kappa n$, so the expected number of errors through time 23 is exactly $46\kappa n$. The count compensator identity in A.2 justifies both expectations. No rare-error premise $\kappa n T \ll 1$ is used.

All proofs in this appendix are conventional finite-process arguments. The original source's finite-semigroup and mission formal evidence is additional, scoped evidence. No new Lean proof compilation was performed for this format edition, and this appendix does not assert that Lean has formalized the full path-space construction or all chemical interfaces.